MySQL Interactive Firmware Development

Documentation

Step 1: Create a Database on MySQL

Ensure that MySQL Workbench is installed and configured. Update the `user` and `password` fields in `mysql_database.py` to match your MySQL Workbench credentials.

Configuration:

```
host="localhost",
user="root",              # Replace with your MySQL username
password="mysql123456",   # Replace with your MySQL password
database="esp32_data"
```

Database Setup:

1. Install the MySQL connector for Python:

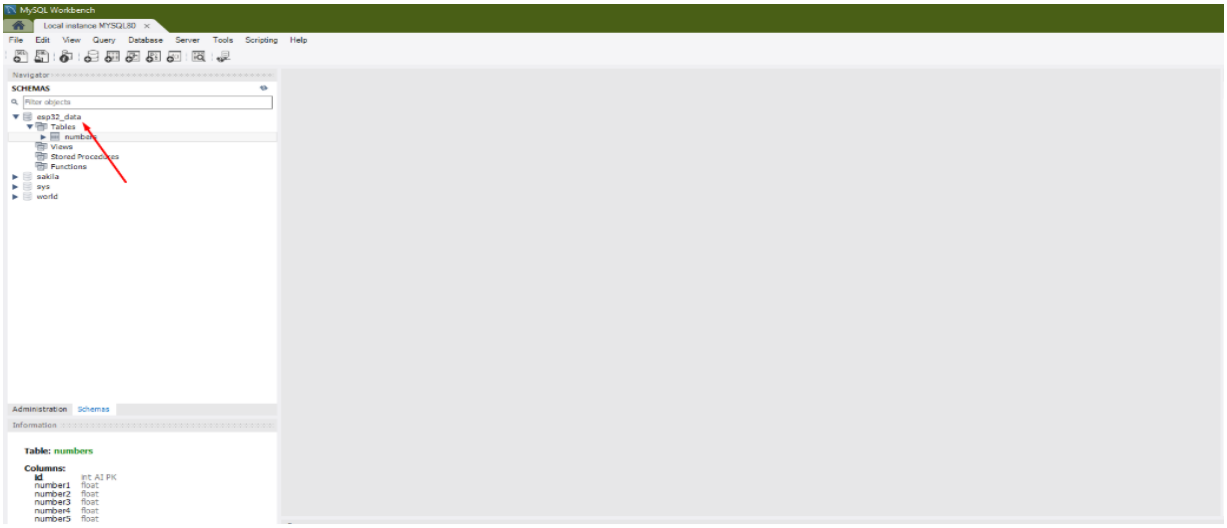
```
pip install mysql-connector-python
```

2. Run the database creation script:

```
python mysql_database.py
```

Verification:

- - Open MySQL Workbench and confirm the creation of the database `esp32_data`.



Step 2: Create a Server

Update the `user` and `password` fields in the server script to match your MySQL Workbench credentials.

Configuration:

```
host="localhost",
user="root",                # Replace with your MySQL username
password="mysql123456",     # Replace with your MySQL password
database="esp32_data"
```

Server Setup:

1. Install Flask:

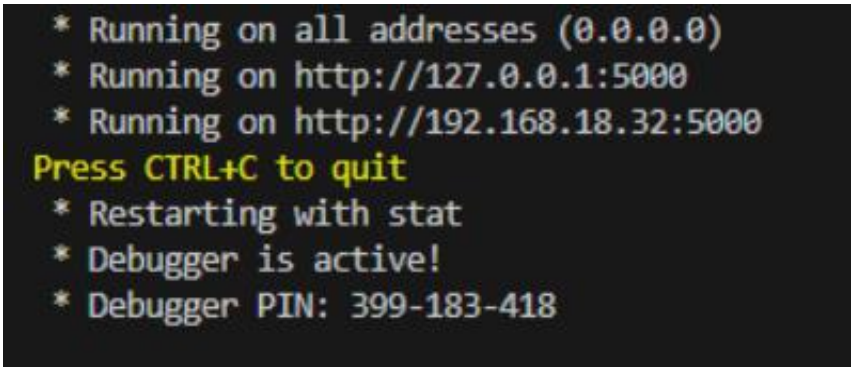
```
pip install flask
```

2. Start the server:

```
python api.py
```

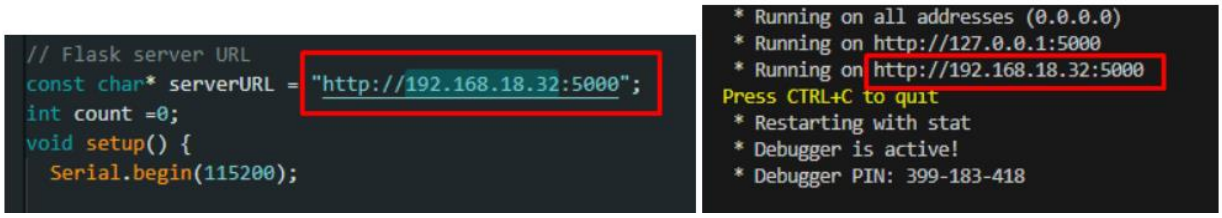
Verification:

- The server should start and display a running message in the terminal.



Step 3: Update Firmware

Ensure that both the server and ESP32 device are connected to the same network.



```
// Flask server URL
const char* serverURL = "http://192.168.18.32:5000";
int count =0;
void setup() {
  Serial.begin(115200);
```

```
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.18.32:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 399-183-418
```

Steps:

1. 1. Compile the firmware using your preferred IDE.
2. 2. Upload the compiled firmware to the ESP32 device.

Note:

- - Confirm network connectivity between the ESP32 and the server.

Resources

Demo Video:

[Watch the Demo Video](https://www.loom.com/share/41fb0520eaf24f73900536a465b07a0f?sid=ff07c38c-4034-4e3b-81e7-210350594474)